BREEZE

# Your Engineering Partner for AI Hardware

From concept to production -- BOM optimization, DFM, prototyping, and manufacturing. China supply chain expertise, global delivery.

| 3,000+ | 12 weeks | 1,000 |
| --- | --- | --- |
| Component suppliers | Concept to prototype | Min production run |

hello@breezehw.com | breezehw.com | Shenzhen, China

No equity. No MOQ surprises. Just engineering.

# Building AI Hardware Is Hard.

Finding the right manufacturing partner shouldn't be.

**80%** of hardware startups fail at the manufacturing stage.

### Factories won't take your 1K MOQ

Most factories want 10K+ units. We start at 1,000.

### Unpredictable NRE costs

A written feasibility sprint gives you NRE, tooling and MOQ quotes before you commit to a build.

### IP leaks and design theft

NDA + IP assignment standard on every engagement. Your designs stay yours.

### Language and timezone barriers

Native English + Mandarin team. We work your hours.

### No DFM feedback until late

We run DFM checks at the design stage, not after you've committed to tooling.

### Certification maze

FCC, CE, UL, SRRC -- we handle the paperwork, testing, and timeline.

# Who's behind Breeze

Four operators. 60+ combined years in Shenzhen supply chain, smart hardware, and cross-border delivery.



## Jett Fu

Founder & CEO

15+ years cross-border operator. Founded AirPop -- smart air wearable launched at CES 2021, covered by Tom's Guide, Wired, Digital Trends. US (Delaware) + HK + Singapore entities. Native EN + ZH. Based in Shenzhen.



## Jank Liu

Supply Chain + Quality Lead

30 years across industrial valves, PCB, wire harness connectors, smart masks, and rail transit. GE Six Sigma Black Belt -- led HX-5 rail NPI. Amphenol Medical cross-border supplier development and compliance lead. Alumni: Tyco, Amphenol, GE.



## Sky

Engineering Lead

10+ years in Shenzhen consumer electronics supply chain. Prototype to mass production for wearables, IoT devices, and audio products. Ground-truth knowledge of the supplier network within 50 km of our office.



## Arting Chen

Partnerships & Deal Flow

AI hardware incubator and investor network. Qualifies deal flow and connects AI hardware founders with Breeze. Direct relationships with YC/HAX-adjacent teams building their first physical product.

# 10+ Years. Not a New Company.

Before Breeze Hardware, our team shipped production at scale:

### AirPop Active+ Halo

World's first smart air wearable -- CES 2021

Bluetooth respiratory sensor + iOS/Android app + certified mask manufacturing. Global retail launch via AirPopHealth direct + national retailers.

**PRESS**

Tom's Guide, Hypebeast, Digital Trends, Wired

### Cross-Border Supply Chain

Customs, logistics, international delivery

Deep fluency in import/export customs clearance, HS-code classification, commercial invoicing, multi-modal freight (ocean, air, rail), bonded warehouse coordination, and destination-country fulfillment into US, EU, LATAM, and APAC.

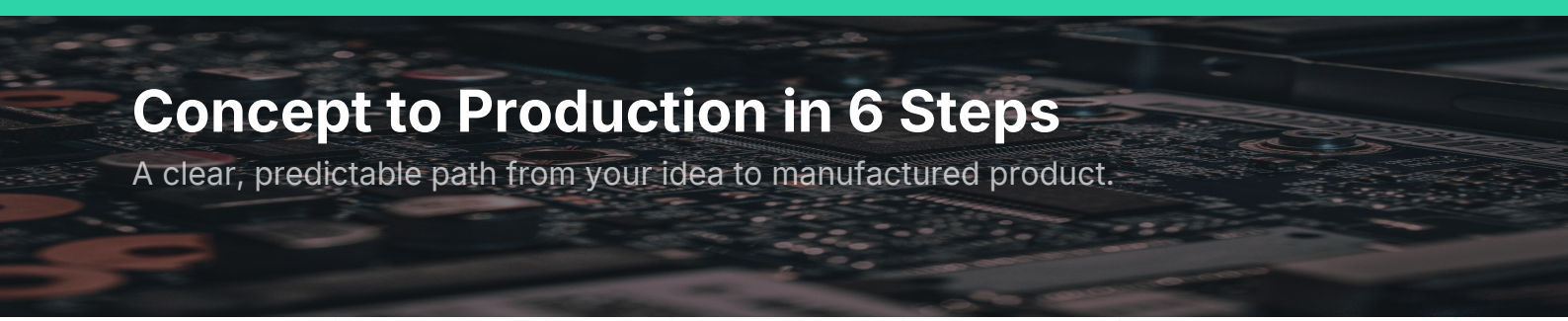
### EU Medical Packaging

Hospital surgical tray systems

Regulated medical device packaging -- material sourcing, sterility validation, third-party lab testing, and full compliance documentation for EU hospital systems.

**Full disclosure.**

Breeze's co-founder Jett Fu is AirPop's founder & CEO. AirPop Halo was delivered by the same supply-chain and engineering team now powering Breeze.



# Concept to Production in 6 Steps

A clear, predictable path from your idea to manufactured product.

01

## Scoping

Share your concept. We assess feasibility, BOM, and ballpark cost.

02

## Engineering Design

Schematic, PCB layout, mechanical design, firmware architecture.

03

## DFM & Prototyping

Optimize for manufacturability. 3-5 functional prototypes.

04

## Tooling & Sourcing

Injection molds, jigs, fixtures. Component sourcing from 3,000+ local suppliers.

05

## Pilot Run

50-500 units. Test assembly line, QC process, packaging.

06

## Mass Production

Scale to 1K, 10K, 100K+. Ongoing QC, logistics, fulfillment.

**12 weeks**
Avg concept-to-prototype

**$26.40/unit**
AI pendant example

**50-500**
Pilot run units

# Cost Reality Check

Real BOM breakdown for a $99 AI wearable pendant (@ 5K units)

**$26.40 / unit** avg landed cost (incl. assembly + packaging)

| Component | Example | Cost Range |
|---|---|---|
| SoC / MCU | ESP32-S3 / nRF5340 | $1.80 - $4.50 |
| Memory (Flash + RAM) | 4MB + 512KB | $0.60 - $1.20 |
| Microphone Array | MEMS x2 | $0.80 - $2.00 |
| Speaker / Haptic | Micro speaker + LRA | $0.50 - $1.50 |
| Battery (LiPo) | 250mAh pouch cell | $1.20 - $2.50 |
| BLE + Wi-Fi Antenna | Chip antenna / FPC | $0.30 - $0.80 |
| PCB (4-layer HDI) | 30x25mm, ENIG finish | $0.80 - $1.50 |
| Enclosure (injection) | PC/ABS, 2-shot mold | $0.60 - $1.20 |
| Connector + Misc | USB-C, pogo pins, ESD | $0.40 - $0.80 |
| PCBA Assembly | SMT + manual + test | $1.50 - $3.00 |
| Packaging | Box, insert, manual, cable | $0.60 - $1.20 |
| **TOTAL (@ 5K units)** | | $9.10 - $20.20 |
| **Landed Unit Cost*** | | $26.40 avg |

* Landed cost includes BOM + PCBA + packaging + testing + 5% yield buffer + freight to US warehouse.
Excludes NRE (one-time engineering), tooling amortization, and certification fees.

# What happens after you sign

Working prototype in 8–12 weeks, pilot batch in 3–6 months, production in 6–12 months. Weekly sync throughout.

**WEEK 1–3**

### Feasibility Sprint

Sign NDA. BOM estimate, factory shortlist with NRE and MOQ quotes, certification path.

**WEEK 4–8**

### Engineering Design

Schematic, PCB layout, mechanical CAD, firmware architecture. Weekly sync.

**WEEK 8–12**

### Prototypes (EVT)

3–5 functional prototypes. DFM checks. EVT gate.

**WEEK 13–20**

### Tooling + DVT

Injection molds, jigs, fixtures. Components from 3,000+ suppliers. DVT units go to certification.

**WEEK 20–26**

### Pilot Run (PVT)

50–500 units. Test assembly line, QC process, packaging.

**MONTH 6–12**

### Production

First production run from 1,000 units, then 10K, 100K+. QC, logistics, fulfillment.

### Defined deliverables at every gate.

EVT, DVT, and PVT gates each have defined deliverables and sign-off. Written status every week. Weekly status sync on Fridays with photos, videos, and next-week plan.

# What It Costs

Cash only. A fixed fee for the sprint, then a monthly fee.

### Feasibility and sourcing sprint
### $3K – $5K

Fixed fee, 2–3 weeks. We check your concept before you commit to a build.

- BOM cost estimate
- Shortlisted factories with NRE, tooling and MOQ quotes
- Certification path (FCC, UL, CE)
- Written risk list and the route we'd take

MOST ENGAGEMENTS

### Prototype to pilot
### $4K – $8K / month

Monthly fee, usually 3–6 months. Engineering to working prototype and pilot batch, with one project manager accountable.

- Schematic, PCB, mechanical, firmware architecture
- 3–5 functional prototypes (EVT / DVT)
- Tooling, jigs, fixtures, owned by you
- 50–500 unit pilot run
- NNN agreements, full source files, written status every week

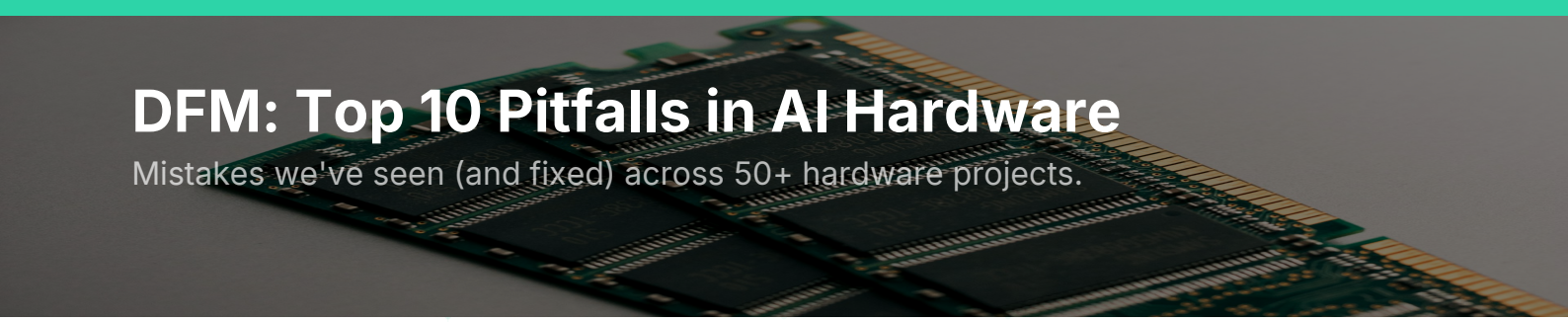
### Production management
### $3K – $6K / month

Monthly fee per product, ongoing. The same project manager runs your production once the pilot passes.

- 1K – 100K+ unit runs
- Inspection planning, QC and yield
- Freight, logistics and replenishment planning
- Production outside China if you need it

**Typical product cost, not our fee: NRE $50K–$150K, $15–$60 per unit at 5K volume.**

Third-party costs (parts, tooling, samples, test labs, specialist engineering) are passed through at cost. Cash only, no equity. After scoping, we can quote a fixed total for the project.



# DFM: Top 10 Pitfalls in AI Hardware

Mistakes we've seen (and fixed) across 50+ hardware projects.

1. **Thermal runaway**
   AI inference chips generate 2-5W. Without proper thermal paths, throttling kills performance.

2. **Antenna detuning**
   The human body and enclosure materials shift antenna performance 3-6 dBm. Simulate with real enclosure.

3. **Battery certification gap**
   UN38.3 + IEC 62133 are mandatory for shipping. Budget 6-8 weeks and $3-5K.

4. **Underspec'd power supply**
   Peak current draw during AI inference can be 3x idle. Size your regulator for bursts.

5. **Acoustic cavity design**
   Mic array performance depends on cavity geometry, not just microphone specs.

6. **Injection mold draft angles**
   Below 1.5 degrees on textured surfaces = stuck parts and scratched finish.

7. **Test jig as afterthought**
   Design the test fixture alongside the product. Retrofitting costs 3x and delays 4 weeks.

8. **Connector placement**
   USB-C orientation and position affect waterproofing, drop strength, and user ergonomics.

9. **ESD protection gaps**
   Every external port needs TVS diodes. Skipping them passes lab testing but fails in the field.

10. **Silkscreen and markings**
    FCC ID, CE mark, and recycling symbols need specific sizes and placements per regulation.

# Certification Roadmap

What you need, how long it takes, and what it costs.

| Certification | Region | Timeline | Budget | Required For |
|---|---|---|---|---|
| FCC Part 15 | United States | 4-6 weeks | $3-8K | Any radio transmitter (BLE, Wi-Fi) |
| CE (RED) | European U... | 6-8 weeks | $5-12K | Radio equipment sold in EU/EFTA |
| UL/IEC 62368 | US / Global | 8-12 weeks | $8-15K | Audio/video/IT equipment safety |
| UN38.3 | Global | 4-6 weeks | $3-5K | Any product with lithium battery |
| IC (ISED) | Canada | 4-6 weeks | $2-4K | Radio devices sold in Canada |
| SRRC + CCC | China | 8-16 weeks | $5-10K | Radio + safety for China market |
| MIC (Giteki) | Japan | 6-10 weeks | $4-8K | Radio equipment sold in Japan |

**Pro Tip**

Start certification pre-compliance testing during the DFM phase.
Finding issues early saves 4-8 weeks and $5-10K in re-spins.
We include pre-compliance checks in our standard engineering process.

# Free Tools for AI Hardware Founders

No signup required. Start estimating your project now.

### 1 BOM Cost Estimator

Input your specs, get a manufacturing cost ballpark in 60 seconds. Covers SoC, memory, sensors, battery, PCB, enclosure, and assembly.

**breezehw.com/tools/bom-estimator**

### 2 DFM Checklist

30-point design-for-manufacturability check covering thermal, mechanical, electrical, and certification pitfalls specific to AI hardware.

**breezehw.com/tools/dfm-checklist**

### 3 NRE Quote Simulator

Estimate your total development cost: engineering hours, tooling, prototyping, certification, and project management.

**breezehw.com/tools/nre-simulator**

### 4 Certification Navigator

FCC, CE, UL, SRRC, UN38.3 -- which do you need? Get a timeline, cost estimate, and testing lab recommendations.

**breezehw.com/tools/cert-navigator**

OPEN SOURCE

**AI Hardware Toolkit** — **github.com/jianjettfu-oss/ai-hardware-toolkit**

Same BOM data, DFM checklists, NRE guidelines we use every day -- free on GitHub.

# Why Breeze?

Four reasons founders choose us over brokers and trading companies.

### Whole Product, Not Just Parts

We build the full device -- electronics, enclosure, firmware, packaging. One team, one responsibility, one timeline.

### No Equity, No Surprises

Cash only: a fixed sprint fee, then a monthly management fee. We don't take a piece of your company.

### Your IP Stays Yours

NDA + IP assignment is standard on every engagement. You receive full source files, schematics, and manufacturing data.

### AI Hardware Native

We understand edge AI thermal management, antenna design for wearables, battery optimization, and low-power ML inference.

**Breeze vs. the alternatives:**

Trading companies add 20-40% margin and zero engineering value.
In-house teams cost $500K+/year and take 6+ months to hire. We're ready now.

# How we compare

Honest differences, not marketing fluff.

| Feature | Breeze | Fictiv / Xometry | HAX / SOSV | Titoma | Trading Co |
|---|---|---|---|---|---|
| **Scope** | **Whole product** | Parts only | Accelerator | Whole product | Sourcing only |
| **Equity taken** | **None** | None | 6–10% | None | None |
| **Minimum order** | **1,000 units** | 1 unit (parts) | Varies | ~1,000 | 10,000+ |
| **Location** | **Shenzhen** | SF + global | SF + Shenzhen | Taiwan | Shenzhen |
| **AI hardware focus** | **Yes** | No | Some | Generalist | No |
| **Price transparency** | **Tools upfront** | Instant quote | Opaque | Quote-based | Opaque |
| **Language** | **Native EN + ZH** | EN | EN | EN | Broken EN |
| **IP policy** | **Assignment std** | Assignment | Varies | Case-by-case | Often unclear |
| **Open-source toolkit** | **Yes (4 tools)** | No | No | No | No |

Breeze is the only whole-product engineering partner in Shenzhen with open-source tools, 1K-unit MOQ, and native bilingual delivery.

# Ready to Start?

1

### Try Our Free Tools

Get a BOM estimate and DFM checklist at breezehw.com/tools. No signup required.

2

### Book a Scoping Call

30 minutes. Share your concept, we assess feasibility and provide a ballpark cost range.

3

### Receive Your Proposal

Scope, timeline and fees in writing: a fixed sprint fee, then a monthly fee.

4

### Start Building

We start with the 2–3 week feasibility and sourcing sprint, then one accountable project manager runs your build.

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Fixed sprint + monthly fee | No equity | Full IP transfer